Responsible use of Large Language Models for Research: LLMs for scientific writing

Lodewijk Pet | Clinical Epidemiology | PhD candidate | l.a.pet@lumc.nl
Bram van Dijk | Public Health and Primary Care | Postdoc | b.m.a.van_dijk@lumc.nl

Course website: https://bma-vandijk.github.io/RULLMR/



Programme

1. In the news: LLMs writing research articles
2. Your experience: using LLMs for scientific writing

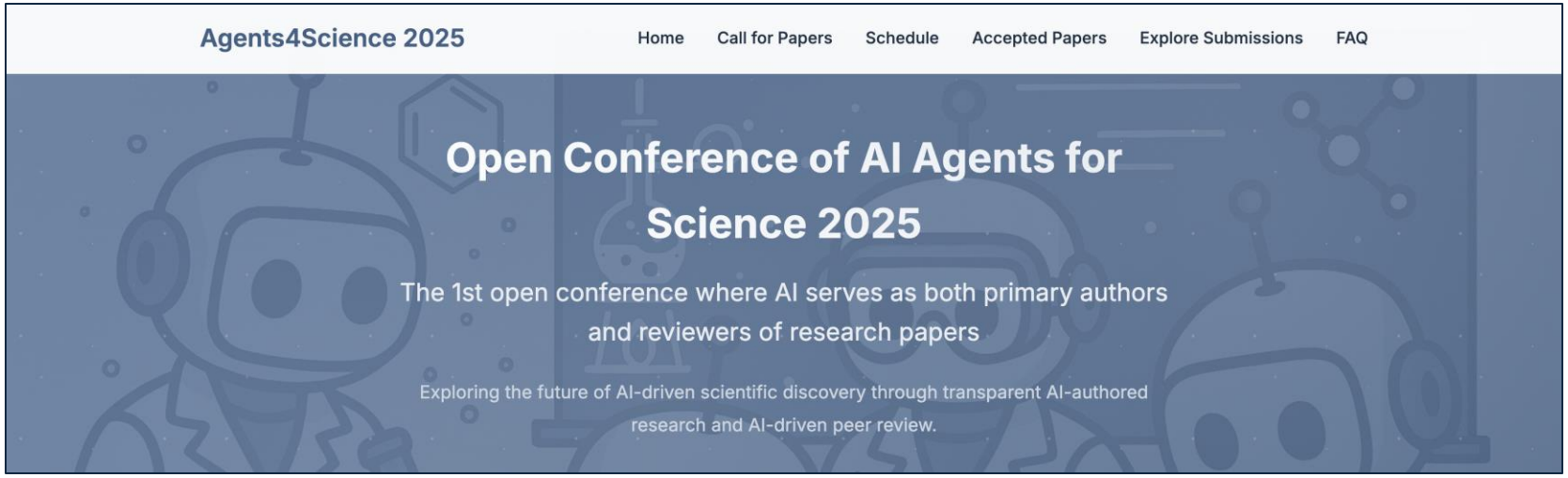
Learning goals

1. Formulate a substantiated opinion on LLM-produced text in research
2. Use LLMs to write, edit or polish your text

1. In the news: LLMs writing research articles

1.1 In the news: LLMs writing articles

- Is this our future?





1.2 In the news: LLMs writing articles

- What is at stake here and who feels at risk?

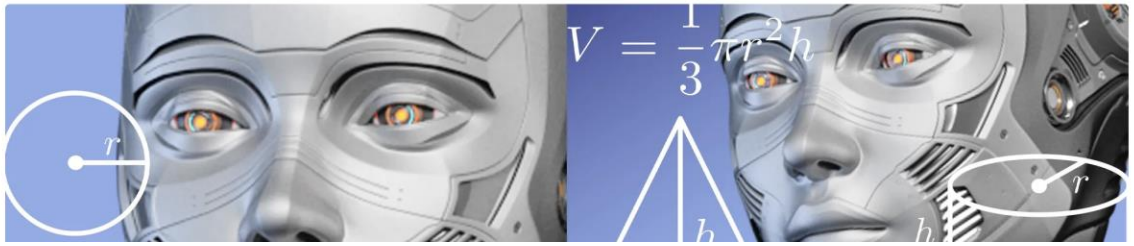
Tech Brew

Stories Topics Resources Events Subscribe

AI

AI math isn't mathing

OpenAI's latest dump of math findings has mathematicians asking what gets lost when AI solves a proof.



Leiden Declaration: AI is challenging the core values of mathematics

OPINION • *02 June 2026*

In the 'Leiden Declaration', an international group of researchers warns that AI is putting fundamental values of the discipline under threat. The rise of AI is forcing mathematics to rethink what makes their field reliable and valuable. This is their proposed way forward.

Five threats AI poses to mathematics

The declaration identifies five threats to the mathematical discipline.

1. Unreliable results
Mathematics is built on rigorous proofs that provide clear understanding. However, AI can produce 'proofs' that look convincing but contain almost invisible errors.

2. Lack of proper attribution and violation of copyright
AI models often produce results without citing the human work they build on. This raises questions about recognition and intellectual property.

3. Dependence and inequality
There is a risk that mathematicians will soon depend on access to the latest proprietary AI technology and expensive computational resources in order to produce competitive results. This leads to inequality between researchers.

4. Overhyping of results
Press releases and blogs often make AI claims without proper scientific scrutiny. This leads to overestimating AI's capabilities and underestimating human contributions.

5. Loss of autonomy
When technical feasibility or commercial interests shape research directions, mathematics risks losing autonomy in setting its research agenda.

1.3 In the news: LLMs writing articles

International Committee of Medical Journal Editors (ICMJE)

- Defining the Role of Authors and Contributors

1. Authorship confers
   - Credit
   - Important academic, social, and financial implications.
2. Authorship also implies
   - responsibility and
   - accountability for published work.

1.4 In the news: LLMs writing articles

Who is an author?

The ICMJE recommends that authorship be based on the following 4 criteria:

1. **Substantial contributions** to the conception or design of the work; or the acquisition, analysis, or interpretation of data for the work; **AND**
2. **Drafting the work or reviewing** it critically for important intellectual content; **AND**
3. **Final approval** of the version to be published; **AND**
4. **Agreement to be accountable** for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

1.5 In the news: LLMs writing articles

Artificial Intelligence @ submission time

- At submission, the journal should require authors to **disclose whether they used artificial intelligence (AI)-assisted technologies** (such as Large Language Models [LLMs], chatbots, or image creators) in the production of submitted work.
- Authors who use such technology should describe, in both the cover letter and the submitted work in the appropriate section if applicable, how they used it.
  - If AI was used for **writing assistance**
  - If AI was used for **data collection**, **analysis**, or **figure generation**

1.6 In the news: LLMs writing articles

Artificial Intelligence @ submission time

- Chatbots should not be listed as authors because they cannot be responsible for the accuracy, integrity, and originality of the work, and these responsibilities are required for authorship
- Therefore, humans are responsible for any submitted material that included the use of AI-assisted technologies.
- Authors should carefully review and edit the result because AI can generate authoritative-sounding output that can be incorrect, incomplete, or biased.

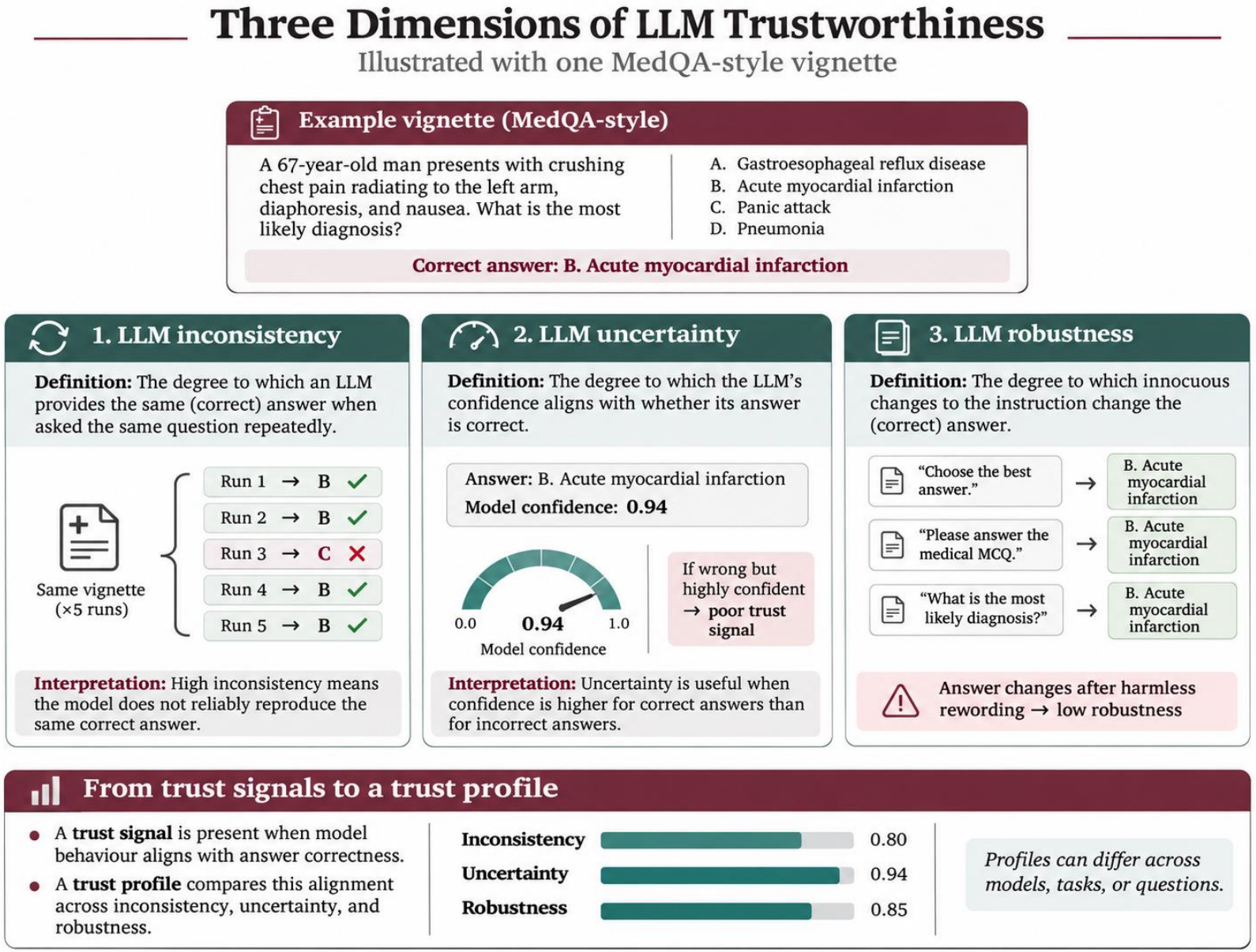
1.7 In the news: LLMs writing articles

Artificial Intelligence @ submission time

- Authors should not list AI and AI-assisted technologies as an author or co-author, nor cite AI as an author.
- Authors should be able to assert that there is no plagiarism in their paper, including in text and images produced by the AI.
- Humans must ensure there is appropriate attribution of all quoted material, including full citations.

2. Your experience: using LLMs for writing

2.1 Your experience: using LLMs for writing (example)



2.1 Your experience: using LLMs for writing (example)

- Let's think together before using AI: what are points in this abstract that could be improved?

Trustworthiness of large language models in medical applications is often evaluated on a single dimension, such as uncertainty or consistency. A systematic literature review of 124 studies identifies inconsistencies in how uncertainty, consistency, and prompt robustness are defined and categorized. We take a different approach by studying these three dimensions together on the MedQA benchmark. For each question, we examine whether the model's behavior aligns with the correctness of its answer across all three dimensions and assign each question a trust profile that summarizes this alignment. We refer to cases where the trust signal is positive but the answer is wrong as hazardous failures. We evaluate two instruction-tuned language models, Mistral-7B-Instruct-v0.2 and Llama-3.1-8B-Instruct, and use the Jaccard index and Cohen's kappa to quantify the overlap of trust failures across dimensions and models. Within each model, the three dimensions overlap partially: some questions fail on multiple dimensions while others fail on only one. Consistency and robustness overlap most, uncertainty and robustness overlap least, and this ordering is the same for both models. About 20\% of Mistral's hazardous failures and 11\% of Llama's occur across all three dimensions simultaneously. Across models, the degree of overlap depends on the dimension: hazardous uncertainty failures are almost entirely model-specific, while hazardous consistency and robustness failures partially overlap between models. Together, these findings show that trust behavior on medical multiple-choice QA is multi-dimensional and that evaluating a single dimension misses failure patterns revealed by joint analysis.

2.2 Your experience: using LLMs for writing (example)

Trustworthiness of large language models in medical applications is often evaluated on a single dimension, such as uncertainty or consistency. A systematic literature review of 124 studies identifies inconsistencies in how uncertainty, consistency, and prompt robustness are defined and categorized. We take a different approach by studying these three dimensions together on the MedQA benchmark. For each question, we examine whether the model's behavior aligns with the correctness of its answer across all three dimensions and assign each question a trust profile that summarizes this alignment. We refer to cases where the trust signal is positive but the answer is wrong as hazardous failures. We evaluate two instruction-tuned language models, Mistral-7B-Instruct-v0.2 and Llama-3.1-8B-Instruct, and use the Jaccard index and Cohen's kappa to quantify the overlap of trust failures across dimensions and models. Within each model, the three dimensions overlap partially: some questions fail on multiple dimensions while others fail on only one. Consistency and robustness overlap most, uncertainty and robustness overlap least, and this ordering is the same for both models. About 20\% of Mistral's hazardous failures and 11\% of Llama's occur across all three dimensions simultaneously. Across models, the degree of overlap depends on the dimension: hazardous uncertainty failures are almost entirely model-specific, while hazardous consistency and robustness failures partially overlap between models. Together, these findings show that trust behavior on medical multiple-choice QA is multi-dimensional and that evaluating a single dimension misses failure patterns revealed by joint analysis.

2.3 Your experience: using LLMs for writing (example)

- We can use Copilot lazily to see whether it aligns with our first thoughts

  Explain concisely in what ways this abstract could | Shared Copilot Chat

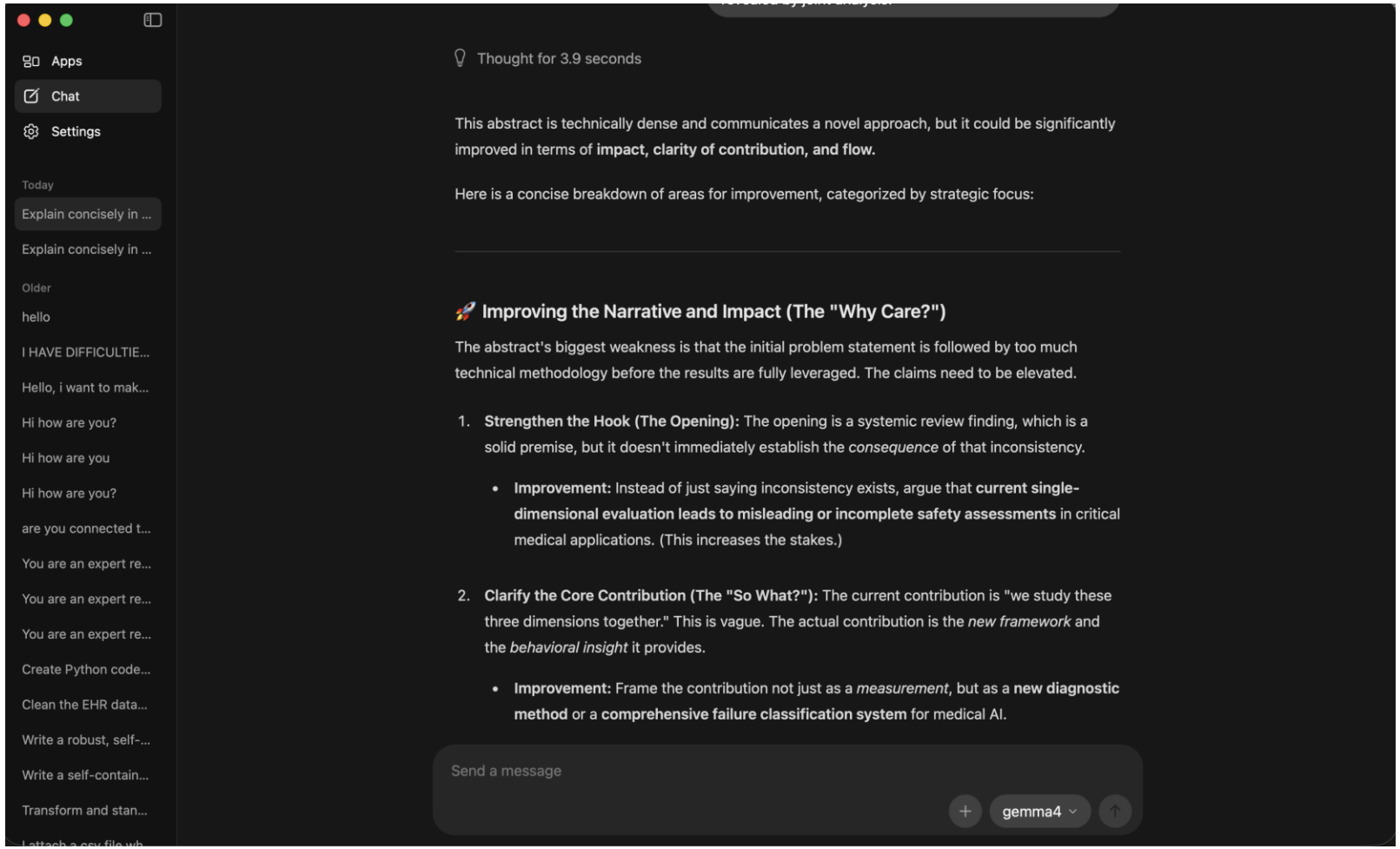
2.4 Your experience: using LLMs for writing (example)

- Would you keep Copilot's abstract as is?

Existing evaluations of the trustworthiness of large language models in medical applications typically assess uncertainty, consistency, or prompt robustness in isolation, making it unclear how these dimensions interact. A systematic literature review of 124 studies identifies inconsistencies in how these dimensions are defined and categorized. To address this gap, we propose a joint evaluation framework that assesses uncertainty, consistency, and prompt robustness simultaneously on the MedQA benchmark. For each question, we categorize whether trust signals and answer correctness align across the three dimensions and assign a trust profile summarizing this relationship. We define cases in which trust signals are positive despite an incorrect answer as hazardous failures. We evaluate two instruction-tuned open-source language models and examine the overlap of hazardous failures within and across trust dimensions and models. Within models, hazardous failures show only partial overlap across dimensions: some questions fail on multiple dimensions, whereas others fail on only one, with consistency and robustness exhibiting the greatest overlap. Across models, overlap patterns differ by dimension: hazardous uncertainty failures are largely model-specific, whereas hazardous consistency and robustness failures are more likely to be shared. Together, these findings demonstrate that trustworthiness in medical multiple-choice question answering is inherently multi-dimensional and that single-dimension evaluations overlook important failure patterns revealed by joint analysis.

2.5 Your experience: using LLMs for writing (example)

- What can local models do?



2.6 Your experience: using LLMs for writing (example)

- Do you like Gemma4's abstract? Why (not)?

Trustworthiness in clinical AI is often assessed using single metrics, such as uncertainty or answer consistency. This narrow focus is fundamentally flawed, as it frequently obscures systemic failure patterns and leads to incomplete safety evaluations. To address this limitation, we introduce a novel, multi-dimensional framework that evaluates the integrated reliability of large language models (LLMs) specifically within medical diagnostic contexts.

We move beyond isolated measurements by analyzing three critical failure dimensions—uncertainty, consistency, and robustness—as a unified "trust profile" for each model answer. This allows us to classify not just if an answer is wrong, but why it fails and what specific dimensions contributed to that failure. Using the MedQA benchmark, we analyze two leading instruction-tuned LLMs (Mistral-7B and Llama-3.1-8B) to identify specific, high-risk failure modes.

Our analysis reveals that model behavior is complex: within each model, failures are not uniform; some questions exhibit singular flaws, while others present multi-faceted weaknesses. Furthermore, we identify crucial differences in how models fail across the dimensions. When comparing models, certain failure types—specifically those related to uncertainty—are highly idiosyncratic to a single LLM. However, failures related to consistency and robustness show significant shared patterns between models, suggesting universal weaknesses in current AI architectures.

These findings demonstrate that relying on a single reliability metric fundamentally misses the full scope of risk. Our joint analysis establishes that an LLM's true performance in clinical settings is a composite feature, requiring multi-dimensional profiling to accurately guarantee safety and pinpoint necessary areas for improvement.

2.7 Your experience: using LLMs for writing

- Who is willing to share task, prompt, and result of using LLMs to do writing? (Homework)

2.7 Please indicate attendance

